

DHYEY PATEL

BACKEND ENGINEER • AI SYSTEMS

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PROFESSIONAL SUMMARY

I design and engineer scalable backend services, secure APIs, and intelligent applications. I focus on bridging the gap between modern AI capabilities and production-ready backend systems. With hands-on experience building AI platforms, enterprise ERPs, and database-intensive SaaS tools, I leverage Python, Django, FastAPI, and PostgreSQL to deliver scalable, reliable backends. I build systems that are clean, maintainable, and built to withstand real-world traffic.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Backend: Django, Flask, FastAPI, REST, JWT

Frontend: REACT, NEXT.JS

Databases: PostgreSQL, MySQL

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, NLP, Pandas, NumPy

Developer Tools: GIT, AWS

PROFESSIONAL EXPERIENCE

Labmentix

Python Development Intern | May 2026 – Present

- Architecting and optimizing production-grade Python backend features and RESTful APIs powering enterprise AI-driven software solutions.
- Collaborating with cross-functional engineering teams to translate abstract system requirements into modular, clean, and test-driven code bases.
- Profiling application runtimes to proactively identify, isolate, and eliminate critical relational database and third-party API processing bottlenecks.

SELECTED ENGINEERING PROJECTS

Finora

Role: Full Stack Engineer

Project Type: Multi-Tenant Cloud ERP

Technology Stack: Next.js, FastAPI, PostgreSQL

Project Links: [Live Demo](#) | [GitHub](#) | [Demo Video](#)

- Architected a multi-tenant cloud ERP with strict double-entry accounting principles, securing isolation via Row-Level Security.
- Implemented an event-driven inventory-to-ledger synchronization engine, ensuring every stock movement generates an immutable journal entry.
- Built a high-throughput backend API utilizing schema validation to reliably process financial payloads.

CodeSentry

Role: Backend & AI Engineer

Project Type: AI-Powered Static Analysis Platform

Technology Stack: React, FastAPI, PostgreSQL, AI

Project Links: [Live Demo](#) | [GitHub](#) | [Demo Video](#)

- Engineered a static analysis platform integrating disparate analysis tools into a unified 5-metric scoring algorithm (Quality, Security, Performance, Readability, Complexity).
- Adopted a decoupled architecture leveraging a backend to asynchronously process code analysis and prompt execution tasks.
- Delivered an enterprise-ready dashboard for intelligent, automated review to identify security vulnerabilities and optimize quality.

Trackademic AI

Role: Computer Vision & AI Backend Engineer

Project Type: Smart Classroom ERP

Technology Stack: FastAPI, Next.js, OpenCV, PostgreSQL

Project Links: [GitHub](#)

- Architected an ERP that processes real-time video streams, achieving 98% facial recognition accuracy in variable lighting.
- Reduced server overhead by offloading heavy image processing to edge nodes, transmitting only lightweight embeddings to the cloud.
- Implemented stateful presence tracking at 15-minute intervals using WebSockets and high-speed vector search with pgvector.

EduFusionERP

Role: Full Stack / Backend Engineer

Project Type: Educational ERP

Technology Stack: Flask, SQLAlchemy, PostgreSQL, JS/CSS

Project Links: [GitHub](#)

- Developed a comprehensive educational ERP serving 5,000+ users, unifying 5 disjointed workflows into a single system.
- Engineered a highly relational database schema in PostgreSQL supporting granular Role-Based Access Control (RBAC) without performance degradation.
- Deployed locally in a test institution, effectively reducing administrative data-entry time by 40% using optimized eager loading queries.

Procura

Role: Full Stack Engineer

Project Type: SaaS Procurement Platform

Technology Stack: Django, React 18, Vite 5, SimpleJWT

Project Links: [GitHub](#)

- Developed a multi-tenant SaaS procurement platform supporting hundreds of distinct organizational hierarchies and approval rulesets.
- Designed a dynamic workflow engine in Django evaluating PO totals against real-time budget thresholds to automatically assign custom sequential and parallel approval steps.
- Implemented strict tenant isolation and audit trails via a heavily constrained PostgreSQL schema and state machine pattern for PO lifecycles.

EDUCATION

Karnavati University

B.Tech in Computer Science | 2024 – 2028

AWARDS

- **1st Place Winner** - Trackademic AI Hackathon (April 2026)